EXP 7: Write a program to implement the Logistic Regression (LR) algorithm in Python

AIM

To implement the Logistic Regression (LR) algorithm in Python and evaluate its performance using a suitable dataset.

SOFTWARE USED

- Google Colab
- Python 3.x
- Pandas Library

PROCEDURE

1. Import the required libraries.
2. Load the Iris dataset.
3. Convert it into a binary classification problem.
4. Split the dataset into training and testing sets.
5. Train the Logistic Regression model.
6. Predict the test data.
7. Calculate and display the accuracy.

GOOGLE COLAB CODE

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target
# Convert to binary classification
y = (y == 0).astype(int)
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)
# Create Logistic Regression model
model = LogisticRegression()
# Train model
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Predicted Values:")
print(y_pred)

print("\nActual Values:")
print(y_test)
print("\nAccuracy:", accuracy)
```

OUTPUT

```
Predicted Values:
[0 1 0 0 0 1 0 0 0 0 0 1 1 1 1 0 0 0 0 0 1 0 1 0 0 0 0 0 1 1 1 1 0 1 1 0 0
 1 1 1 0 0 0 1 1]

Actual Values:
[0 1 0 0 0 1 0 0 0 0 0 1 1 1 1 0 0 0 0 0 1 0 1 0 0 0 0 0 1 1 1 1 0 1 1 0 0
 1 1 1 0 0 0 1 1]


Accuracy: 1.0
```

RESULT

Thus, the Logistic Regression (LR) algorithm was successfully implemented in Python using the Iris dataset, and the model achieved high classification accuracy.